

LexTax AI: A Business Compliance Intelligence Engine (BCIE) for Unified Legal, gst and Vendor Risk Aggregation

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I. INTRODUCTION

Abstract—Information related to compliance within a small or medium enterprise is often dispersed across separate systems: legal contracts are saved as documents, gst invoices are handled via accounting software, vendor records are monitored informally or not tracked at all, and deadlines for contract renewals or filings are maintained in spreadsheets. Many of the existing tools for checking compliance, including recently developed tools that combine large language models with rule-based validation, operate on a single document or a single check. As a result, a risk identified in one area is rarely evaluated in the context of other areas. In this work, we introduce LexTax AI. It functions as a prototype Business Compliance Intelligence Platform, driven primarily by a core Business Compliance Intelligence Engine (BCIE). The system manages risk by linking six distinct operational modules. These handle legal document classification, specific clause analysis, and gst liability prediction. We also integrated tools to catch duplicate gst invoices, validate Input Tax Credit (ITC) eligibility, profile vendor risks, and track ongoing contract lifecycles. The engine combines these signals into an overall explainable business level compliance score and prioritized course of actions. We make no claims of novelty in document classification or invoice validation. The study is about combining the risk scores of multiple modules at the level of a business and reports finding interdependencies between the risk factors that are not evident at the level of individual checks. The prototype was tested on three aspects across the implemented modules: correctness of implementation, consistency of the signal propagation.

Keywords—*Business Compliance Intelligence, Risk Aggregation, gst Compliance, Legal Document Analytics, Explainable AI, Vendor Risk Scoring, Contract Lifecycle Management.*

A major gap in the literature is that existing methods are rarely linked to one another into a single compliance picture. the specific research gap. Earlier research has looked into automating compliance checks for unstructured business files by relying on language models and orchestration setups [1]. Within legal text mining, legalbert proved that training a model directly on legal datasets significantly sharpens its understanding of that specific domain [2]. Along similar lines, the cuad framework offers a structured way to pinpoint contract clauses needed for targeted legal review [3]. While these models excel at their isolated jobs, they fall short when it comes to tying those findings together. Specifically, they fail to link legal text to broader enterprise metrics like gst data, vendor details, contract lifecycles, or impending deadlines. Beyond legal text, researchers have built tools for other pieces of the corporate compliance puzzle. For example, Sentence-BERT transforms text blocks into semantic vectors so we can easily compare their underlying meanings [4]. To handle these vectors efficiently, libraries like faiss enable fast storage and approximate nearest-neighbor searches across massive embedding collections [5]. When dealing with scanned invoices, extraction tools like PaddleOCR pull out the necessary data fields [9]. From there, simple rule-based logic can step in to figure out the applicable gst. As for forecasting overall gst liability, tree-based algorithms—especially Random Forest—perform exceptionally well on tabular data structures [6]. In fact, these models have already proven highly effective for direct gst prediction tasks [7]. For contract lifecycle

management, previous work has shown that large pretrained language models can be used to identify contract events, as well as predict time-to-event [8]. By analogy, one could identify relevant deadlines and predict their timing in contract documents. For gst liability prediction, well-established tree-based methods such as Random Forest [6] and XGBoost [7] are suitable choices for predictive modelling. SHAP, an explainability method, also demonstrates why the user needs to see what factors drove the model’s output [8]. However, these techniques were developed for individual analytical tasks.

They are not designed to aggregate different compliance findings into a business-level assessment. The gst component also needs to be based on statutory and administrative requirements, gst Council recommendations and associated regulatory guidance form the basis for checking tax conditions and calculations [10]. For this reason, LexTax AI separates the extraction of information from documents and invoices from the deterministic compliance validation. In doing so, a gst finding can be traced back to the specific rules and extracted values that it is based on, rather than relying on a predictive model whose reasoning is not visible. The gap the work addresses is the one that exists in current ability for integrating different kinds of compliance information into unified, explainable business-level risk assessment. Each of the available methods are useful tools for the legal NLP, semantic similarity, OCR, prediction or explainability tasks, but they do not tackle the problem of assessing whether the risk signals they detect can be aggregated in some manner that allows for joint consideration on a common scale and a transparent aggregation layer, with each result traceable to its input components.

Figure I contrasts the existing manual process with the LexTax AI platform along the dimensions this work targets. The comparison is structural rather than a performance benchmark, since no performance claim is made until Section VII is populated with measured results.

Aspect	Existing Manual Process	LexTax AI Platform
Tools required	Word (drafting) + Tally (GST) + Excel (tracking) + external lawyer review	Single unified platform
Legal document review	Manual, lawyer-dependent, slow to turn around	AI-assisted drafting and rule-based clause/missing-clause review (professional review still recommended)
GST validation	Manual calculation, error-prone	Deterministic rule engine, checked at entry time
Vendor risk visibility	Not tracked systematically	Aggregated, persistent vendor risk profile
Cross-source risk view	Fragmented across tools, no linkage between them	Unified BCIE compliance score with ranked, explained reasons
Historical trend tracking	Manual spreadsheet upkeep	Persisted, queryable history per business

Fig. 1. Existing Manual Process vs. LexTax AI Platform

The comparison in Figure I summarizes the structural differences between the conventional fragmented compliance process and the proposed LexTax AI platform. The comparison focuses on the dimensions addressed by this work, including the number of tools involved, legal document review, gst validation, vendor risk visibility, cross-source risk assessment, and historical trend tracking. The table is intended to describe the workflow-level distinction between the two approaches and is not presented as a quantitative performance benchmark. The

motivating scenario in this paper shows why information from different sources has to be considered together. Suppose a gst invoice from a vendor, ABC Suppliers, contains a 12

II. RELATED WORK

The work on legal document analysis has leaned towards domain-specific language models and annotated contract datasets. Training models specifically on legal corpora, as seen with LEGAL-BERT, clearly boosts results for tasks like text classification and data extraction [2]. On the evaluation side, CUAD provides a heavily annotated dataset built by experts to help review contract clauses [3]. While both tools are great for automating legal analysis, they are mostly limited to scanning individual documents. They don’t bridge the gap to connect those text-based insights with wider enterprise compliance data.

To handle the matching side of the proposed system, we rely on semantic similarity. Sentence-BERT is highly effective at creating and comparing text embeddings [4], and FAISS steps in to run fast similarity searches across huge vector databases [5]. Combining them makes it easy to spot duplicate or overlapping records. However, just finding duplicates doesn’t give a complete picture of overall business compliance.

Handling invoices usually breaks down into pulling the text and then validating the rules. Tools like PaddleOCR extract the raw data from scanned images [9], allowing us to run explicit checks for tax discrepancies afterward. For the forecasting side, algorithms like Random Forest [6] and XGBoost [7] remain the go-to choices for making predictions on structured data. Yet again, these are isolated solutions. None of them merge legal risks, gst data, vendor profiles, and deadlines into one unified platform. Finally, while recent work has tested large language models to run compliance checks on unstructured corporate files [1], explainability frameworks like SHAP have become essential to actually show how these models arrive at their decisions [8]. LexTax AI considers all these as building blocks of its approach. Its innovation lies in the combination of the different outputs: the outputs of each of the modules are normalized into risk signals and combined through the explicit, traceable BCIE aggregation layer.

III. SYSTEM DESIGN AND ARCHITECTURE

A. Overall Architecture

LexTax AI is a compliance-intelligence platform built from separate modules, and each module is responsible for analyzing one type of compliance information of business data in order to provide standardized risk signals. The Business Compliance Intelligence Engine (BCIE) aggregates the results of all these modules’ work, performing an integrated compliance assessment of the company’s performance. Document control, legal entity management, predictive compliance, and risk consolidation modules are separated and, therefore, can be analyzed individually, remaining integral parts of the unified assessment.

B. A. Overall Architecture

The framework is made up of six main components: legal document analysis, gst invoice validation, gst liability prediction, vendor risk assessment, contract lifecycle management, and compliance deadline tracking. Every module retains its distinct processing logic while generating a structured output that features a normalized risk signal along with the contributing factors. B B transactional transactional transactional transactional transactional transactional transactional transactional transactional transactional documents. This division allows for the modification or substitution of particular components without altering the assembly procedure.

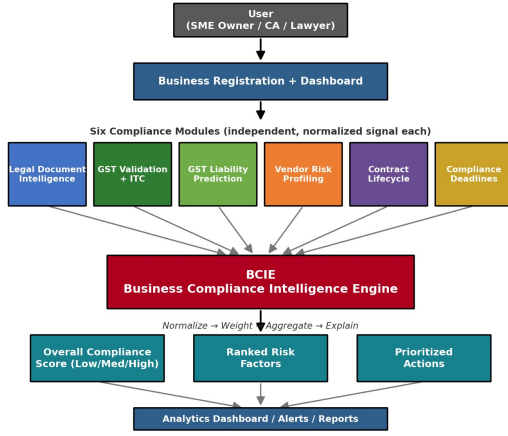


Fig. 2. the system architecture diagram

The overall architecture is illustrated in Fig. 2. The workflow begins with business and document information entering the appropriate processing module. Legal documents are analyzed for document type, clauses, missing provisions, and possible contradictions, while gst invoices undergo field extraction and deterministic validation. Historical financial records are processed by the prediction module, and vendor, contract, and deadline information is converted into corresponding risk indicators. Every module feeds its results directly into the BCIE. The engine then merges these separate signals to build one complete final assessment.

C. B. Legal Document Intelligence Module

The legal document module works with four types of agreement: employment agreements, non-disclosure agreements, vendor agreements, and service agreements. Files that are uploaded undergo processing for extracting text and classifying types, followed by clause segmentation and validation of anticipated clauses. Each kind of document is associated with a checklist of expected provisions, allowing the module to identify potentially missing clauses. The module additionally conducts a rule-based verification of contradictions among clauses concerning related topics. For instance, contradictory termination or notice-period clauses may be identified as possible inconsistencies that need human or professional

legal examination. The system also looks for duplicate or near-identical documents. It converts text into sentence-level embeddings and compares them with a similarity search. The module generates a uniform legal-risk indicator along with the identified missing clauses, potential contradictions, and additional supporting evidence.

D. C. gst Compliance Module

The gst module deals with structured invoice entries as well as with scanned or image invoices. The system uses OCR to read key details from each invoice, including the supplier and customer gstIN, taxable value, gst rate, Cgst, Sgst, Igst, and HSN/SAC codes. The gst calculations are carried out with fixed validation rules, not a learned model. Recalculation of the expected tax amount from the taxable value and rate and comparison with the values indicated on the invoice are made. Arithmetics with a known correct answer are not given to the learned model. For finding duplicates in invoices, two checks are made: an exact match of the relevant fields and a similarity check. In case of a discrepancy that may affect the Input Tax Credit, the module highlights the potentially affected amount with an indication for review rather than a definitive assessment.

E. D. gst Liability Prediction Module

The gst liability prediction module utilizes the past monthly financial records of sales, purchases, output gst, input gst, Input Tax Credit, and tax paid to the government. Before the data goes into the predictive model, it is turned into three kinds of input: past tax values, month-to-month changes, and seasonal categories. Linear Regression, Random Forest, and XGBoost can all be trained and evaluated against a common held-out test set. For the final forecasting, we simply select the model with the lowest error. Forecasting a liability is a probabilistic task compared to the deterministic procedure of statutory gst validation since it relies on historical trends to estimate future liabilities rather than a mathematical formula.

F. E. Vendor Risk Module

The vendor risk module keeps a history for each vendor based on transaction-level indicators, such as number of invoices, past gst errors, incidents of duplicate-invoice and submission delays in documents. The system combines these indicators into a single vendor-level risk signal by following a fixed, defined procedure. The module also captures causes that pushed up the risk score, and the BCIE can therefore refer to these drivers in explaining a result. A vendor with recurring gst errors, or high numbers of duplicate invoices, is therefore likely to see that its vendor risk signal has increased. If we have enough transaction data, we could easily plug in anomaly detection as an extra layer of analysis. However, regardless of the data volume, the prototype will always generate its final score using transparent, straightforward metrics.

Component	Technique	Input Type	Output Type
Legal document classifier	Legal-BERT (fine-tuned transformer)	Extracted document text	Document type + confidence
Clause similarity / duplicate detector	Sentence-Transformers + FAISS	Document / clause text	Similarity score + matched sections
OCR extraction	PaddleOCR / EasyOCR	Scanned image / PDF	Structured text fields
GST validation	Deterministic rule engine (no ML)	Structured invoice fields	Pass / mismatch flag
GST liability predictor	XGBoost (compared against Linear Regression, Random Forest)	Monthly financial time series	Predicted next-month liability
Vendor risk / anomaly scorer	Rule-based score + Isolation Forest	Vendor transaction history	Risk score + anomaly flag

Fig. 3. AI/ML Components Summary

G. F. Contract Lifecycle and Deadline Modules

The contract lifecycle module stores the contract’s metadata (i.e. vendor, start date, expiry date, and renewal period) to determine the contract status based on the date-based logic. The contract statuses include Created, Active, Expiring soon, Renewal/Review, and Expired. The deadline module performs a similar function for the compliance activities. In other words, it stores the filing’s type, due date, and recurrence and determines the filing’s urgency. The same principle of two modules implies that the fewer days remaining to the date of expiry/renewal/due date, the higher is the risk/urgency level. The outputs of the modules are the status, remaining days (to the date) and the normalized risk/urgency signal.

H. G. BCIE Aggregation Layer

The BCIE sits at the center of the platform. It collects the results from every module and uses them to judge the overall compliance status of a business. Every module produces both a numerical signal and explanatory information describing the reasons for them. BCIE handles neither the original legal documents nor the invoices. It does not replace the specific models and rules used by every module. It provides a common view over heterogeneous compliance findings and allows for aggregating and comparing different results. The mathematical formulation of the normalization of the input values, weighting, aggregation, risk bands, and explanation mechanism are detailed separately in Section IV.

I. H. Technology Stack

The prototype was built with React and Tailwind CSS, and TypeScript for the frontend and Python, with FastAPI as the backend framework. PostgreSQL was utilized for relational data storage and management, including business, vendor, contract, legal-document, and gst-invoice data, and object storage was used for storing uploaded and generated documents. Depending on the module, the system also draws on legal-text processing, OCR, semantic embeddings, similarity search, and machine-learning models.

By design, the architecture strictly isolates standard rule based tasks from the trained predictive models. gst calculations and validation rules, duplicate checks on particular fields,

and date calculations for contracts and deadlines all take place independently of machine learning. Machine-learning and language-processing algorithms are applied only where they are required for the task: classification, semantic analysis, similarity estimations, and predictions.

J. End-to-End Workflow

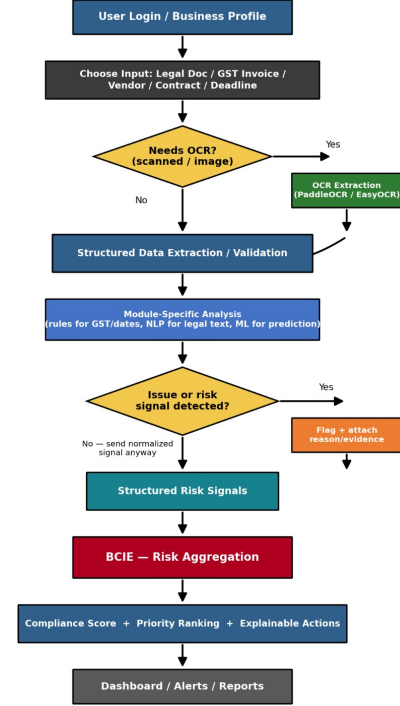


Fig. 4. the end-to-end workflow flowchart:

The entire processing flow is presented in Fig. 3. First, the user inputs the business data, legal documents, invoices, past financial statements, or other applicable information. Second, depending on the input type, it is directed to the appropriate module for processing. The scanning and OCR step is required only if the input was a scanned file. Each module produces a structured output containing its normalized risk signal and supporting factors. These outputs are passed to BCIE, including outputs indicating that no issue was detected. BCIE then aggregates the signals and generates the business-level compliance assessment, contributing factors, and corresponding explanations for presentation through the dashboard. In this architecture, each compliance analysis is carried out on its own, and the business-level risk assessment is handled in a separate step on top of them. Because of this setup, we can evaluate each module independently. The BCIE then acts as the central hub, merging these scattered compliance signals into one transparent, trackable final score.

IV. BCIE RISK AGGREGATION MODEL

The Business Compliance Intelligence Engine (BCIE) provides the integration layer between the specialized compliance modules and the final business-level assessment. Each module produces a normalized risk signal in the range [0,1] where 0 indicates no detected risk and 1 indicates the maximum risk represented by that module. Normalization is required because the module outputs may have different forms and units. As an example, the legal module might produce a clause-risk value, while the contract lifecycle module reports how many days are left before a contract expires. For signals that are not naturally expressed as risk probabilities, BCIE applies a fixed normalization function before aggregation. In the case of contract and deadline urgency, the remaining time is converted into a bounded risk value that increases as the due date or expiry date approaches. Because every module's output is scaled to the same numerical range, the BCIE can combine them consistently.

The overall business-level risk is calculated using a weighted-sum model:

$$R_{\text{overall}} = w_1 S_{\text{legal}} + w_2 S_{\text{gst}} + w_3 S_{\text{vendor}} + w_4 S_{\text{contract}} + w_5 S_{\text{deadline}} \quad (1)$$

where S_{legal} , S_{gst} , S_{vendor} , S_{contract} , and S_{deadline} represent the normalized signals from the corresponding modules. The weights are configuration parameters and satisfy $\sum_{i=1}^5 w_i = 1$.

The initial configuration used in the prototype is $w_1 = 0.20$ for legal risk, $w_2 = 0.30$ for gst risk, $w_3 = 0.25$ for vendor risk, $w_4 = 0.15$ for contract risk, and $w_5 = 0.10$ for deadline urgency. These values represent the documented configuration used for the prototype and are not learned from historical compliance outcomes. Therefore, they should be interpreted as design parameters rather than statistically validated estimates of relative compliance importance.

The resulting R_{overall} value is mapped to predefined risk bands. Under the current prototype configuration, a value of 0.70 or above is classed as High risk, and anything below that falls into the Medium or Low band categories according to the configured thresholds.

The weighted contribution of module i is calculated as

$$C_i = w_i S_i. \quad (2)$$

The contributions are ranked in order to identify those with the most impact to the overall assessment. BCIE retains the underlying explanation generated by the corresponding module. Rather than just outputting a standalone risk score, the platform explains the exact reasoning behind that number. For example: 'Vendor ABC showed gst errors in 6 of their last 50 invoices' This allows the user to trace the aggregate assessment back to the operational evidence that generated the individual signal.

The aggregation layer is intentionally deterministic; BCIE does not apply an additional learned model after the specialized modules have calculated their outputs. In this way,

the final risk calculation can be reproduced directly from the module signals and configured weights while retaining the source of each contribution visible to the reviewer. The approach then emphasizes traceability and inspectability of the aggregation process. The effect of the configured weights and the limitations of their validation are further discussed in Section X.

The application of this aggregation model to a concrete vendor scenario is described in Section V.

V. ILLUSTRATIVE CALCULATIONS

This section demonstrates the operation of the BCIE aggregation model and the deterministic gst validation logic using representative numerical examples.

A. BCIE Weighted Risk Score

For the Vendor ABC scenario, the normalized signals are $S_{\text{legal}} = 0.55$, $S_{\text{gst}} = 0.82$, $S_{\text{vendor}} = 0.78$, $S_{\text{contract}} = 0.70$, and $S_{\text{deadline}} = 0.65$. Using the weights defined in Section IV,

$$R_{\text{overall}} = (0.20)(0.55) + (0.30)(0.82) + (0.25)(0.78) + (0.15)(0.70) + (0.10)(0.65) = 0.721. \quad (3)$$

The resulting risk value of 0.721 falls within the current High-risk band. The corresponding Business Compliance Score is $100(1 - 0.721) = 27.9$. The weighted contributions are retained separately, allowing the final assessment to be traced to the individual module signals.

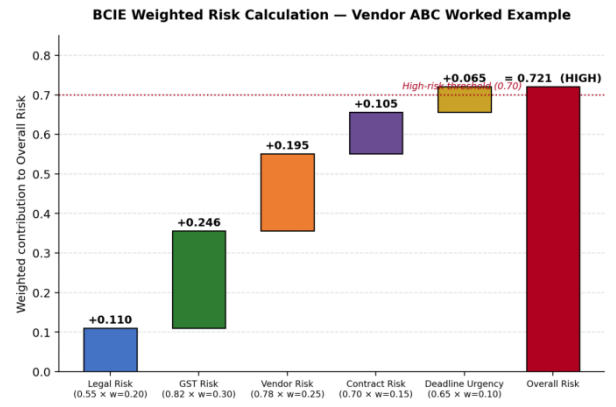


Fig. 5. Weighted contributions to the Vendor ABC overall risk score.

B. gst Rate Validation

The gst module relies on fixed arithmetic rules to check tax rates and amounts. For an invoice with a taxable value of 80,000 and an applicable 18% rate, the expected gst is

$$\text{gst} = 80,000 \times 0.18 = 14,400. \quad (4)$$

For an intra-state supply, this corresponds to $C_{\text{gst}} = 7,200$ and $S_{\text{gst}} = 7,200$. If the state 28% rate is applied, the

calculated gst becomes 22,400, producing an 8,000 difference. The module flags this condition as a rate mismatch.

These examples demonstrate the two principal functions of the framework: BCIE combines heterogeneous risk signals into a traceable business-level assessment, while deterministic gst rules provide directly verifiable statutory calculations.

VI. EXPERIMENTAL METHODOLOGY

We evaluated the prototype at two levels: how well each module performs on its own, and how well the modules work together once they are connected through the BCIE. Module-level evaluation ensures that each module delivers the expected output to accomplish its task. End-to-end evaluation ensures that the BCIE correctly synthesizes, normalizes, and explains the multiple outputs coming from the different modules. This separation allows the contribution of the proposed architecture to be evaluated independently from the performance of the individual algorithms used within the modules.

A. Experimental Setup

The evaluation was conducted using the prototype implementation described in Section III. The software environment consists of Python-based backend services, FastAPI, PostgreSQL, FAISS, OCR and natural-language processing components, and machine-learning libraries used by the individual modules. The experiment uses constructed compliance data because access to a sufficiently large anonymized enterprise compliance corpus was not available. The test data were therefore designed to reproduce representative compliance conditions while maintaining known expected outcomes for evaluation.

B. Test Data and Evaluation Scope

The legal-document testing procedure makes use of publicly available template agreements, as well as synthetically created documents with controlled modifications, such as removal of mandatory clauses, introduction of contradictions between related provisions, and near-duplicates. The supported document classes include employment, non-disclosure, vendor, and service agreements. gst evaluation makes use of constructed invoice records which contain both valid and deliberately corrupted cases. The test cases included gst rate mismatches, incorrect Cgst, Sgst, and Igst splits, duplicate and near-duplicate invoices, and invoices with missing details. The gst liability prediction experiment makes use of monthly historical records which contain sales, purchases, output gst, input gst, Input Tax Credit, and tax-paid values. We introduced a synthetic temporal variability to the data, thus obtaining temporal patterns of repeated growth and seasonality. For the vendor, contract, and deadline evaluations, we used handwritten records, which covered a variety of both risk levels and urgency constraints.

C. Compliance Test Scenarios

The evaluation is organized into eleven structured scenarios covering the principal functions of the prototype. The scenarios include classification of legal documents, identification of

missing clauses and contradictions between the contracts, document similarity check, calculation of gst and validation of tax-split, finding exact and near duplicate invoices, identification of ITC impact, gst liability forecasting, vendor risk assessment and urgency check for contracts and deadlines. One end-to-end BCIE scenario is included, which contains various risk conditions. For the end-to-end scenario, the system receives a combination of gst, vendor, and contract signals corresponding to a high-risk business condition. The BCIE score, risk band, weighted contributions, and stated reasons are then checked against the results expected from the configured model.

D. Evaluation Metrics

Classification tasks, including document-type classification and vendor-risk classification, are evaluated using accuracy, precision, recall, and F1-score. Detection tasks, including missing-clause identification, contradiction detection, document similarity, and duplicate-invoice detection, are evaluated using precision and recall against the known set of planted or predefined test conditions. gst calculation and tax-split validation are evaluated as deterministic correctness checks. A test case is considered correct when the computed tax values and applicable Cgst/Sgst/Igst allocation agree with the expected rule-based result. We evaluated duplicate-invoice screening on cases where the exact and near-duplicate invoices were already known. We evaluated gst liability prediction with three error measures: mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE). Linear Regression, Random Forest, and XGBoost are evaluated under the same time-series validation procedure, allowing their prediction errors to be compared under a common experimental setting. BCIE is evaluated separately from the module-level prediction metrics. The aggregation calculation is verified by recomputing the weighted score from the recorded module signals. Explanation consistency is assessed by checking whether the highest-ranked reported reasons correspond to the module contributions that actually determine the aggregate score.

E. Experimental Procedure

Each test case is processed through the corresponding module using the same configured rules and model versions throughout the evaluation. The predicted or detected output is compared with the predefined expected outcome, and the resulting metric is recorded. For BCIE evaluation, the module outputs are stored before aggregation so that the final score can be independently recomputed and its individual contributions traced. No modification of the expected labels is performed based on the obtained results. The purpose of the evaluation is to determine whether the implemented system behaves consistently with its predefined rules, test conditions, and expected outcomes. The resulting measurements are reported in Section VII.

VII. RESULTS AND EVALUATION

The prototype was evaluated on classification, detection, gst prediction, and BCIE aggregation tasks. The document

classification module reached 100% accuracy: missing-clause detection had a recall of 95.4%. Duplicate-invoice detection was also found to have 100% accuracy while the vendor-risk scoring was the weakest result with 57.3% accuracy.

Task	Accuracy	Precision	Recall	F1
Document classification	100.0%	100.0%	100.0%	100.0%
Missing-clause detection	—	28.3%	95.4%	43.7%
Contradiction detection	77.5%	100.0%	35.7%	52.6%
Document similarity	81.0%	64.2%	100.0%	78.2%
Duplicate invoice detection	100.0%	100.0%	100.0%	100.0%
Vendor risk scoring	57.3%	38.2%	41.3%	39.7%

Fig. 6. Classification and detection results.

For gst liability prediction, Linear Regression achieved the lowest reported error with an MAE of 4,356 and MAPE of 4.74%. BCIE aggregation was verified using the Vendor ABC scenario, producing $R_{\text{overall}} = 0.721$ and a Business Compliance Score of 27.9.

Model	MAE (₹)	RMSE (₹)	MAPE
Linear Regression	4,356	5,402	4.74%
Random Forest	5,504	6,616	6.15%
XGBoost	6,613	7,806	7.39%

Fig. 7. gst liability prediction results.

The results show that performance varies across compliance tasks. Deterministic duplicate and document-classification checks achieved high scores, while clause, contradiction, and vendor-risk tasks showed greater variation. This indicates that the reliability of the overall compliance assessment depends on both the quality of individual module outputs and the transparent aggregation performed by BCIE.

VIII. DISCUSSION

Our testing proves that rule-based checks and machine learning models can run smoothly within one architecture. We rely on deterministic logic for gst calculations, contract expiries, and statutory deadlines since those tasks have clear, absolute answers. Conversely, we reserve machine learning for areas needing pattern recognition, like document classification, semantic matching, and predicting gst liability. The BCIE brings these outputs together using a straightforward weighted-sum formula, letting us easily track each module’s impact on the final score. That said, this simple linear approach is a limitation because static weights cannot capture the complex, overlapping relationships between various risk factors. Upgrading to learned aggregation could solve this, but it adds significant complexity. That route really only makes sense once you have enough real-world training data from an actual business environment. As of now, the architecture covers legal, gst, vendor, contract, and deadline risks, while normalized scores of each module contribute to the business-level assessment. Therefore, the approach could be expanded to include other areas within a company, such as payroll liabilities, licensing, and workplace safety. The results also indicate that a unified compliance architecture should not require every module to use the same analytical technique. Deterministic validation is

appropriate where expected outcomes are explicitly defined, while learned methods are useful for classification, semantic comparison, and forecasting. BCIE provides the common interface between these heterogeneous components.

IX. DESIGN IMPLICATIONS AND CONSIDERATIONS

A. Deterministic and Learned Components

LexTax AI separates deterministic compliance operations from learned analytical tasks. We rely on hardcoded logic to manage standard checks like gst calculations, tax splits, contract expiries, and deadlines. Meanwhile, the more complex features such as document classification, semantic matching, and predicting gst liability depend entirely on machine learning and embedding models. This separation keeps rule based results directly verifiable.

B. Explainability

BCIE retains individual module signals and their weighted contributions, allowing the overall risk assessment to be traced to its underlying factors. This provides reviewers with the specific causes associated with a reported risk rather than only a final numerical value.

C. Module Independence

Each module communicates with BCIE through a standardized normalized signal and supporting information. This allows individual models or processing components to be modified or replaced without changing the core aggregation mechanism and supports the addition of future compliance modules.

D. Data and Regulatory Dependency

Vendor-risk scoring and gst liability prediction require substantial quantities of historical data to be reasonably informative, while gst validation depends on current tax rates. For these reasons, when there is insufficient historical data, such signals are not very useful. On the other hand, any changes in gst rules necessitate a continuous update of the underlying rules in the validation process.

E. Cross-Module Integration Analysis

The module-level results show that the errors differ from one compliance task to another. High results in deterministic validation and duplicate detection indicate that rule-based checks can be evaluated directly against predefined conditions, whereas clause, contradiction, and vendor-risk tasks remain more dependent on the characteristics of the underlying data. These differences are a good reason to separate each specialized analysis from the BCIE’s aggregation layer. At the aggregate level, the BCIE takes the various forms of output from each module and turns it into a single representation. In the Vendor ABC scenario, the combined signals produce $R_{\text{overall}} = 0.721$, with gst and vendor risk contributing 0.246 and 0.195, respectively. The contribution values can be recomputed directly from the configured weights and module signals, allowing the final assessment to be traced to its sources. This

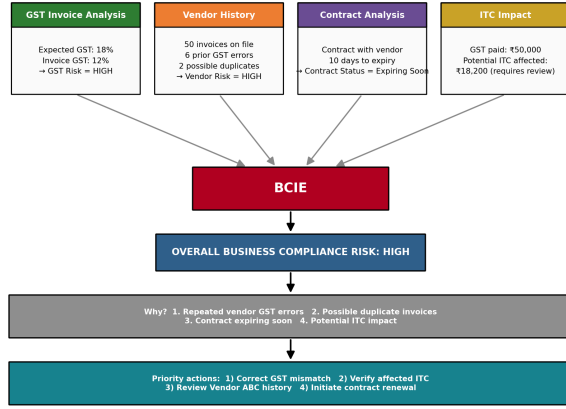


Fig. 8. BCIE end-to-end Vendor ABC risk assessment and explanation.

demonstrates the intended relationship between module-level analysis and business-level compliance assessment without treating the aggregate value as a learned probability.

X. LIMITATIONS

The reported results represent a prototype evaluated using synthetic and template-derived data rather than real business records at production scale. Legal evaluation covered a limited set of document types and constructed test cases, so performance on other document types, jurisdictions, and drafting styles remains untested. gst validation used fabricated invoices, while gst liability prediction used a synthetic monthly time series, which may not fully represent the variability of real business data. Vendor-risk scoring is rule-based and has not been validated against actual audit outcomes; therefore, it should be treated as a risk-indication mechanism rather than a predictor of future compliance failures. Similarly, legal clause and contradiction detection is intended to support document review, not provide legal advice. Drafts produced by the system, along with any issues it flags, should be checked by a qualified professional before anyone relies on them in practice.

XI. CONCLUSION AND FUTURE WORK

This paper described the LexTax AI prototype Business Compliance Intelligence Platform utilizing the BCIE as a transparent aggregation layer, which accumulated signals from the legal, gst, vendor, contract, and deadline modules, and distilled them down to the business-level compliance assessment. The value of this prototype was largely determined by its architectural novelty, which underpinned the combined application of the tools used. The signals processed by the system underwent normalization and were aggregated into a final business-level compliance assessment using a deterministic weighted model, the parameters of which remained available for inspection.

The value of such an approach was primarily determined by the fact that the business level compliance assessment used

different signals that originated from the separate analyses performed by the system. In the case of deterministic checks, such as gst and deadlines, the processing of these signals yielded results that could be interpreted as standalone compliance assessments. At the same time, analyzing the legal, vendor, and gst forecasts gave us a much deeper context for evaluating the business as a whole. We specifically designed the architecture to separate raw compliance detection from actual decision making. Each module has one core job: flagging specific issues like tax mismatches, absent clauses, duplicate files, or tight deadlines. From there, the BCIE steps in to consolidate everything. Keeping these steps distinct means we preserve the evidence for every single flag, rather than just spitting out a generic, unexplainable final score. Because of this, our project serves purely as an advisory tool. It is built to speed up human reviews, not replace human judgment or make autonomous choices.

Looking ahead, we have three primary goals for future development. First, we want to test learned aggregation methods on real audit data to see how they hold up in practice. Second, we plan to refine our vendor risk models as we get our hands on more labeled historical records. Finally, we need to implement a versioned gst database to ensure our rules always match the latest tax regulations. Since this is currently a prototype, our immediate next step involves rigorous testing against larger, anonymized corporate datasets alongside extensive user feedback to truly validate its real world effectiveness.

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